ITM University Gwalior

School of Computing and Intelligent Systems

Project-Based Learning (PBL)

Mini Project Report — Submission Format

Course: Data Analytics Using Python

Course Code: CSL0312

Program: B.Tech. CSE (Artificial Intelligence & Machine Learning)

Semester: III Section:

Academic Session: July–Dec 2026

Project Details

Project Title: Disease Frequency Analysis using Healthcare Records

Domain : Healthcare

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Date of Submission: 12/10/2026

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1. Introduction & Problem Statement

- The project **"Disease Frequency Analysis Using Indian Healthcare Records with Python"** analyses a healthcare dataset obtained from Kaggle using Python-based data analysis techniques.
- Healthcare data analysis helps identify patterns in recorded patient information and understand how disease records vary across demographic and clinical categories.
- The dataset initially contained 600 records and 19 attributes. After data cleaning and preprocessing 590 records were retained for analysis. The project uses Pandas, NumPy, Matplotlib and Seaborn to clean, process, analyse and visualize healthcare data.
- The main objectives are to analyse disease frequency, compare diagnoses by gender and age group, examine regional distribution, study BMI and treatment outcomes, compare Blood Glucose and HbA1c levels, investigate numerical correlations and explore monthly record frequency.
- The problem addressed is how to transform healthcare records into meaningful information through data cleaning, statistical summaries and visualizations.
- Power BI and Streamlit are planned for developing an interactive dashboard and a Python-based web application to present the findings.

2. Dataset Description & Source

Dataset Name: Indian Healthcare Patient Records

Source / Organization: Kaggle Datasets

Dataset Link (URL):https://www.kaggle.com/datasets/arunsworkspace/indian-healthcare-patient-records

Type of Dataset: ☐**Public** ☐**Private** ☐**Self-Collected**

Number of Records:590 after cleaning (600 originally selected)

Number of Attributes/Columns:19

☐ Dataset meets minimum quality criteria (Section 7)

Brief description of attributes/columns:

The dataset contains numerical as well as text-based categorical columns. The numerical columns include Age, BloodGlucose, HbA1c_%, TotalCholesterol, and BMI, which help analyse patients' age, blood sugar levels, average HbA1c, cholesterol levels, and body mass index. The text-based columns include Gender, Region, SocioeconomicStatus, Occupation, Symptoms, PrimaryDiagnosis, TreatmentType, TreatmentOutcome, Imaging Type, ImagingFindings, HospitalType, and InsuranceCovered. These columns help classify patients based on their characteristics, diseases, treatment details, and healthcare facilities. PatientID is used as a unique identifier, while VisitDate helps analyse monthly disease frequency and changes in patient visits over time. Overall, these attributes help identify the most frequently recorded diseases, compare disease records across age groups and genders, study regional patterns, examine treatment outcomes, and explore relationships between clinical indicators.

3. Data Cleaning & Preprocessing

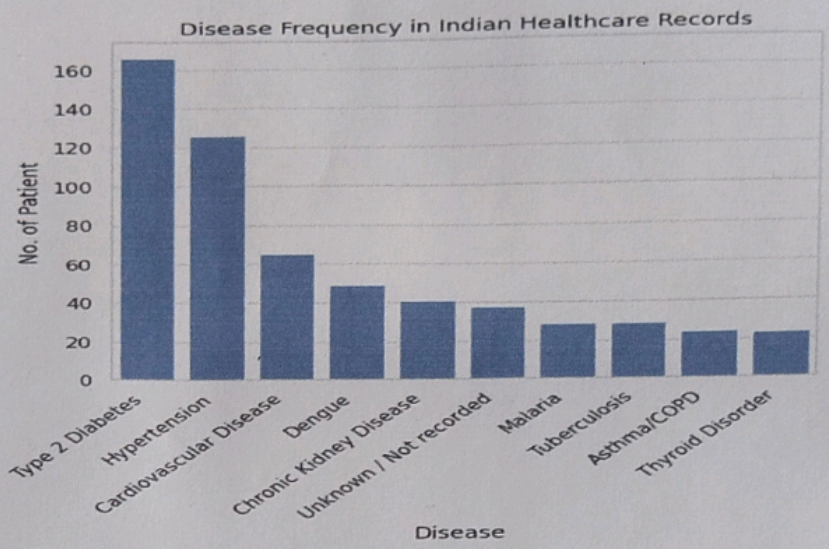
- **Duplicate Removal:** Initially, the dataset contained 600 records. After removing 10 duplicate records, the final dataset contained 590 records and 19 columns.
- **Handling Missing Values:** Missing values were identified in columns such as HbA1c_%, BMI, TreatmentOutcome, and ImagingType. Appropriate methods were used to handle these missing values.
- **Handling Invalid Values:** The value 999 in the Blood_Glucose_mg_dL column was identified as an invalid placeholder. These values were replaced with missing values and then filled using the median.
- **Missing Value Imputation:** Missing HbA1c and BMI values were handled using median-based imputation, while missing treatment outcomes were labelled as "Unknown".
- **Standardization of Text Data:** Variations in disease names under PrimaryDiagnosis and entries in the Gender column were standardized to maintain consistency.
- **Handling Imaging Data:** Missing ImagingType values were filled with "No imaging performed" wherever the corresponding imaging findings indicated that no imaging had been performed.
- **Data Type Conversion:** The VisitDate column was converted into a proper datetime format to support monthly and time-based analysis.
- **Column Correction:** An incorrectly named column, Bloo_Glucose_mg_dL, was removed during preprocessing.
- **Outlier Analysis:**The Interquartile Range (IQR) method was used to detect potential outliers in numerical columns such as Age, BMI, Blood_Glucose_mg_dL, HbA1c_%, and Total_Cholesterol_mg_dL. First, Q1 (25th percentile) and Q3 (75th percentile) were calculated, followed by IQR = Q3 − Q1. The lower and upper limits were determined using the formulas Q1 − 1.5 × IQR and Q3 + 1.5 × IQR, respectively. Values outside these limits were flagged as potential outliers and examined carefully. Plausible values were retained to preserve meaningful patient data, while the invalid blood glucose value 999, found in four records, was replaced with missing values and imputed using the median value of 122.1.
- **Final Dataset:** After cleaning and preprocessing, the dataset contained 590 records and 19 columns and was ready for exploratory data analysis, statistical analysis, and visualization.

4. Exploratory Data Analysis (EDA)

- **Disease Frequency Analysis:** The frequency of different diseases was analysed to identify the most commonly recorded diseases in the dataset. Type 2 Diabetes had the highest number of records, followed by Hypertension and Cardiovascular Disease.
- **Gender-Based Analysis:** Disease records were grouped according to gender to compare the distribution of different diseases among male and female patients.
- **Age Group Analysis:** Patients were divided into age groups such as 0–18, 19–35, 36–50, 51–65, and 66+. Disease frequencies were compared across these groups to understand the distribution of disease records by age.
- **Regional Analysis:** Patient records from different regions were analysed to compare the number of cases recorded for diseases such as Type 2 Diabetes and Hypertension.
- **BMI-Based Analysis:** Body Mass Index (BMI) values were categorized into Underweight, Normal, Overweight, and Obese groups. Disease records were compared across these categories to explore patterns between BMI categories and recorded diseases.
- **Treatment Outcome Analysis:** Diseases were compared with their treatment outcomes to examine the distribution of recorded outcomes for different diagnoses.
- **Clinical Indicator Analysis:** Average blood glucose and HbA1c levels were calculated for different diseases. Type 2 Diabetes showed the highest average blood glucose level (184.12 mg/dL) and average HbA1c level (8.15%) among the analysed disease groups.
- **Correlation Analysis:** Correlation analysis was performed on numerical variables, including Age, BMI, blood glucose, HbA1c, and total cholesterol. Blood glucose and HbA1c showed a positive correlation of approximately 0.434.
- **Monthly Trend Analysis:** Patient records were grouped by month using the Visit_Date column. July 2024 had the highest overall monthly record count, with 34 records, while Type 2 Diabetes had its highest monthly count in May 2025, with 13 records.
- **Purpose of EDA:** The analysis helped summarize disease frequencies, compare patient groups, examine clinical measurements, and identify patterns for further visualization and interpretation.

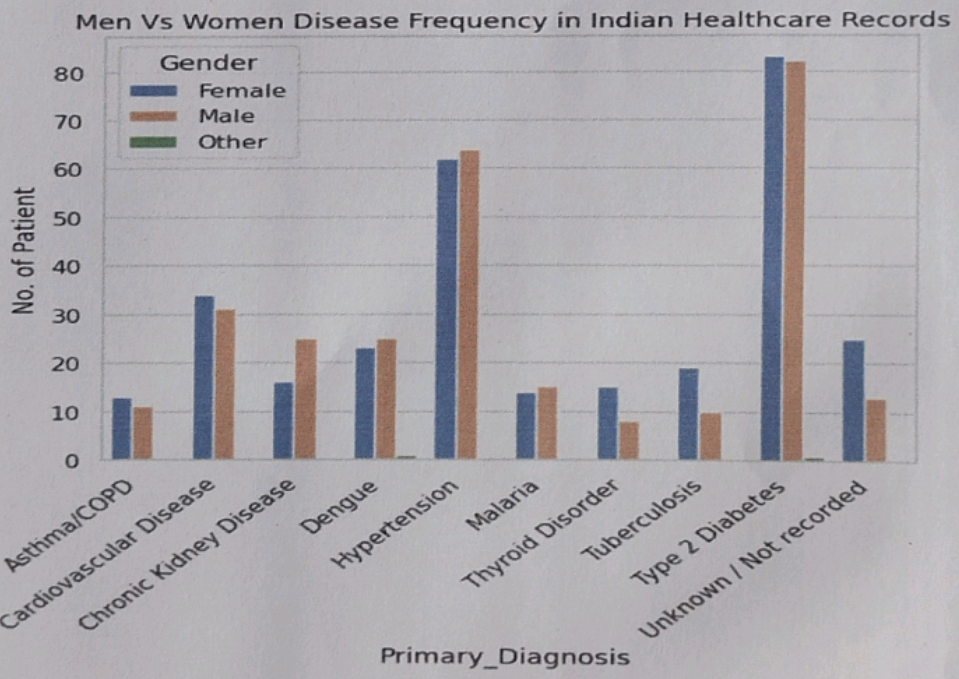
5. Data Visualization & Observations

5.1 Disease Frequency Analysis



Observation :-Type 2 Diabetes has the highest frequency in the dataset, followed by Hypertension and Cardiovascular Disease. Asthma/COPD and Thyroid Disorder have comparatively fewer records. This shows which diagnoses occur most frequently in the available dataset.

5.2 Disease Frequency by Gender



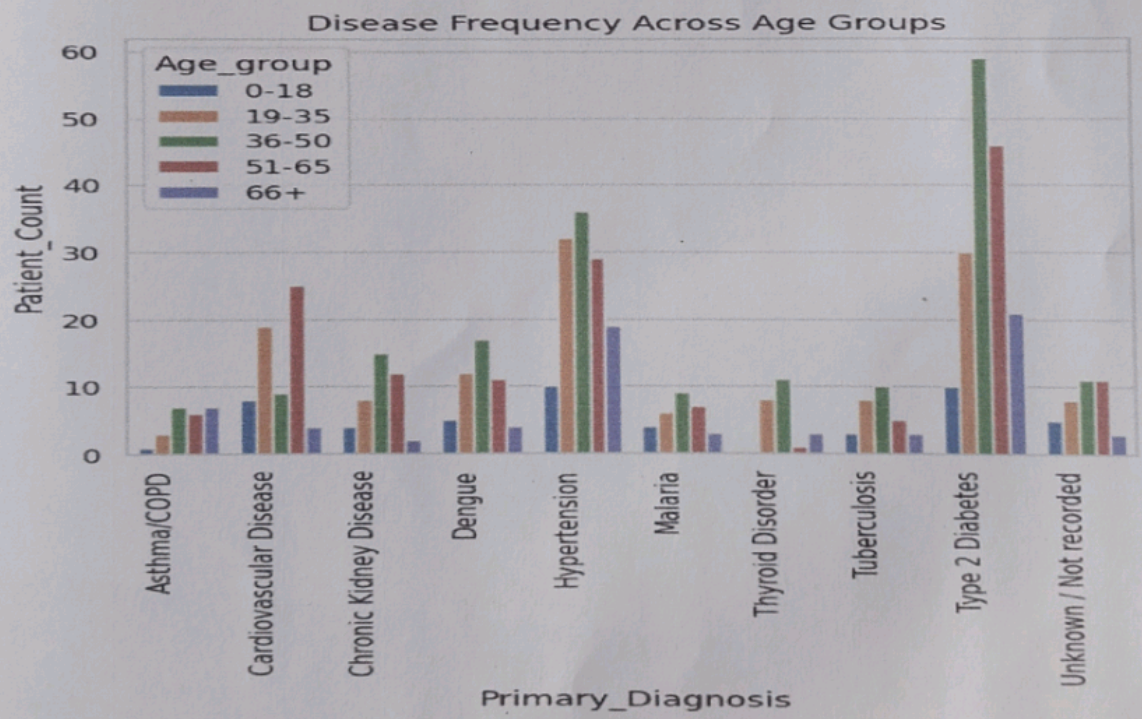
Observation:-The chart compares disease frequency among female, male, and other gender categories. Type 2 Diabetes and Hypertension have high record counts for both females and males, while differences between genders vary across diagnoses.

5.3Type 2 diabetes Geographic Distribution Across India



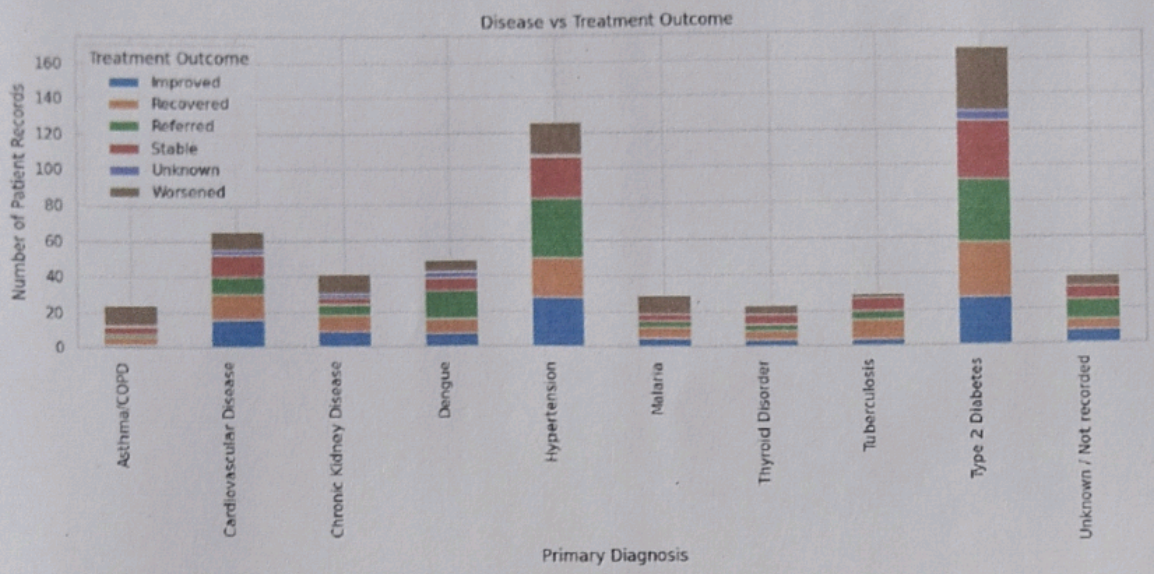
Observation :-The choropleth map shows the regional distribution of recorded Type 2 Diabetes patients. The color intensity varies across the displayed regions, indicating differences in the number of recorded patients. Darker shades represent higher patient counts, while lighter shades represent lower counts. These results reflect the available dataset and do **not** necessarily indicate the actual prevalence of diabetes in each region.

5.4Disease Frequency Across Age Groups



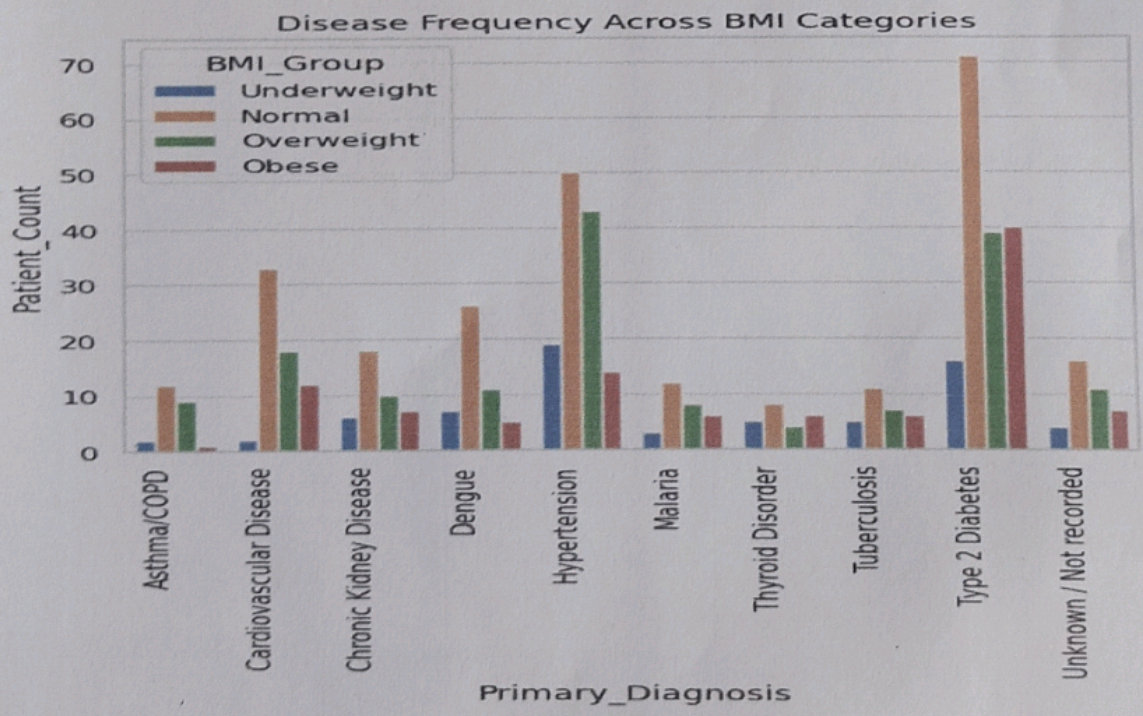
Observation :-The 36–50 age group has the highest number of Type 2 Diabetes records, followed by the 51–65 age group. Hypertension is also frequently recorded among adults. These patterns describe the dataset and do not establish age as the cause of any disease.

5.5 Disease vs. Treatment Outcomes



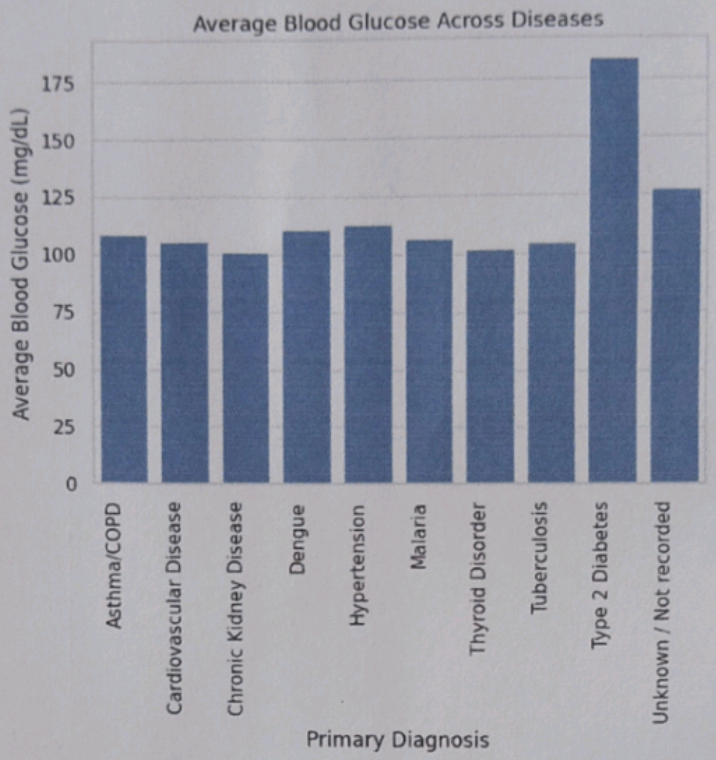
Observation :-The stacked bar chart compares treatment outcomes across different diagnoses. Type 2 Diabetes and Hypertension have the largest total numbers of records. The chart helps compare outcome categories, although the number of records alone does not indicate which disease has the best treatment success rate.

5.6 Disease Frequency Across BMI Categories



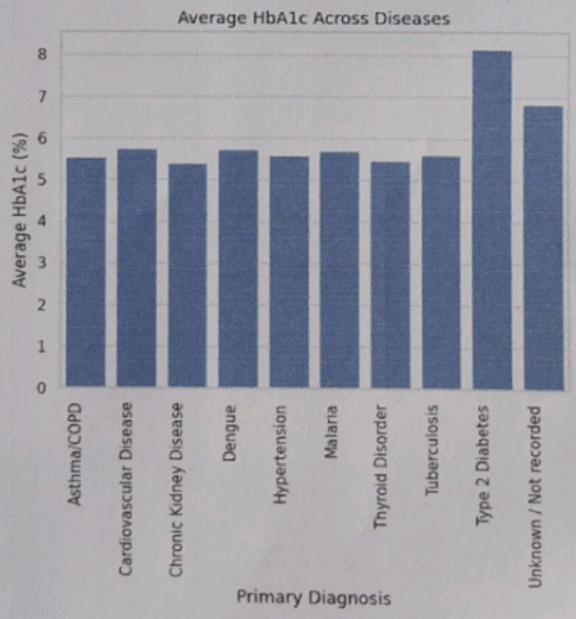
Observation :-The Normal BMI category contains the highest number of records for several diagnoses, including Type 2 Diabetes and Hypertension. Overweight and Obese categories also contain substantial numbers of records for some diagnoses. These are frequency comparisons, not evidence of a causal relationship between BMI and disease.

5.7 Average Blood Glucose Across Diseases



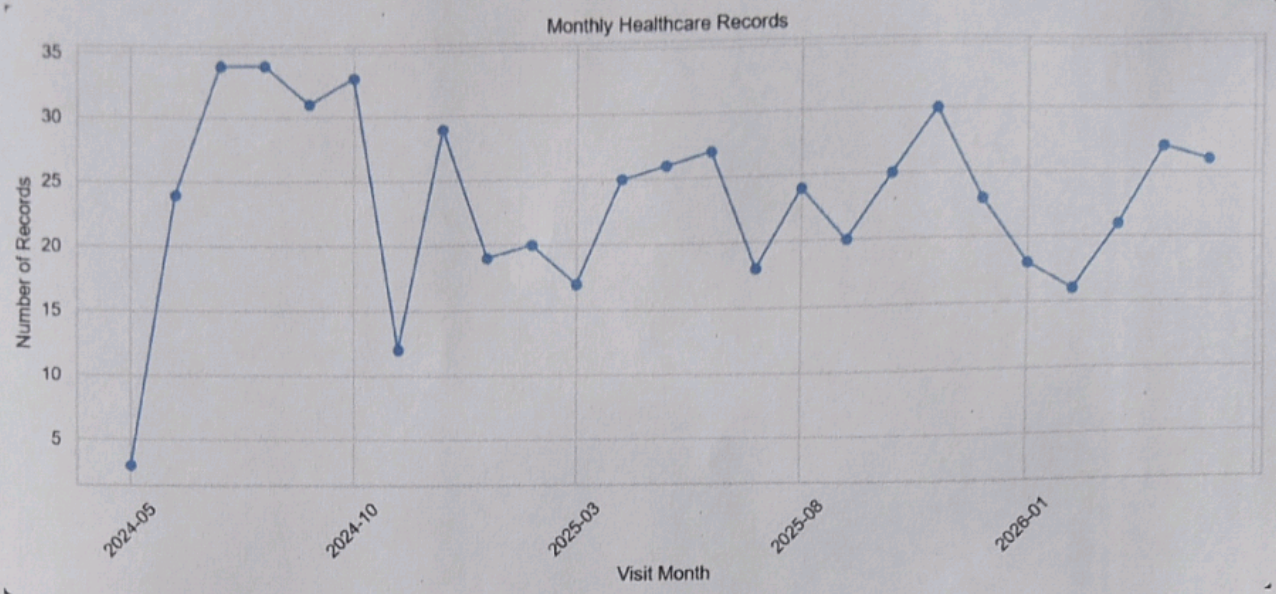
Observation :-Type 2 Diabetes has the highest average blood glucose level, at approximately 184 mg/dL. The other recorded diagnoses have average values mostly around 100–113 mg/dL, while the Unknown / Not recorded category is approximately 126 mg/dL. These averages describe the dataset and should not be used alone for medical diagnosis.

5.8 Average HbA1c Across Diseases



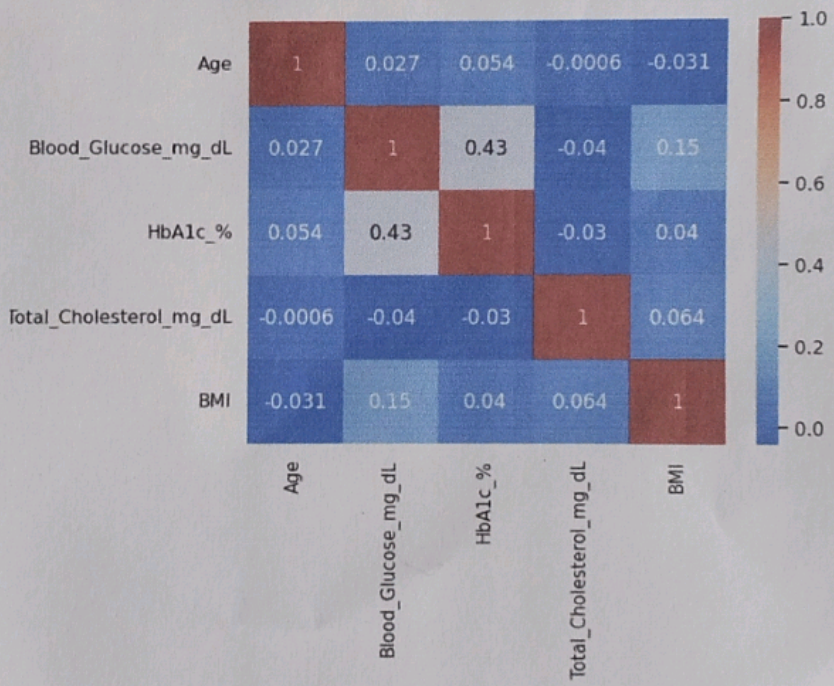
Observation :-Type 2 Diabetes has the highest average HbA1c level, at approximately 8.1%. The other diagnosis categories generally have average values between 5.4% and 5.7%, while Unknown / Not recorded is approximately 6.8%. This indicates a difference in average HbA1c levels across the recorded diagnosis categories.

5.9 Monthly Healthcare Record Trends



Observation :-The line chart shows that the number of healthcare records varies across months. Some months have more than 30 records, while others have fewer than 20. These changes reflect the distribution of records over time and do not necessarily represent changes in actual disease incidence.

5.10Correlation Heatmap of Clinical Indicators



Observation :-Blood Glucose and HbA1c show a moderate positive correlation of approximately 0.43. Blood Glucose and BMI have a weak positive correlation of approximately 0.15. Most other variable pairs show correlations close to zero, indicating weak linear relationships in this dataset.

6. Statistical Analysis (Basic)

- **Descriptive Statistics:** Basic statistical analysis was performed on numerical columns such as Age, BMI, Blood_Glucose_mg_dL, HbA1c_%, and Total_Cholesterol_mg_dL to understand the distribution of patient data.
- **Disease Frequency Count:** The frequency of each disease was calculated using grouping and counting methods. Type 2 Diabetes had 177 records, followed by Hypertension with 134 records and Cardiovascular Disease with 68 records.
- **Mean Calculation:** The mean values of blood glucose and HbA1c were calculated for different disease categories to compare their average clinical measurements.
- **Median Imputation:** Median values were used to handle missing numerical data, including missing HbA1c and BMI values. Invalid blood glucose values were also replaced with the median after being identified as missing.
- **Correlation Analysis:** A correlation matrix was calculated for selected numerical variables. Blood glucose and HbA1c showed a positive correlation of approximately 0.434, indicating that higher values of one variable tended to be associated with higher values of the other in this dataset.
- **Group-Based Analysis:** Grouping and aggregation techniques were used to compare disease records by gender, age group, BMI category, treatment outcome, and month of visit.
- **Interpretation of Results:** The statistical analysis helped summarize disease frequencies, compare average clinical measurements, and identify relationships between numerical variables. These findings are descriptive and do not establish causation or represent disease prevalence in the entire Indian population.
- **Tools Used:** Pandas and NumPy were used for data grouping, counting, statistical calculations, and correlation analysis.

7. Key Insights & Interpretation

1. **Disease Frequency Analysis:** Type 2 Diabetes was the most frequently recorded disease with 177 records, followed by Hypertension with 134 records.
2. **Gender-Wise Analysis:** Type 2 Diabetes records were nearly equal among female (83) and male patients (82), while some other diseases showed differences in gender-wise counts.
3. **Regional Analysis:** Tamil Nadu had the highest recorded Type 2 Diabetes count (25), followed by Karnataka (24). These figures represent records in the dataset, not actual regional prevalence.
4. **Age Group Analysis:** Type 2 Diabetes and Hypertension had comparatively high record counts in the 36–50 age group, while Cardiovascular Disease was relatively more represented in the 51–65 age group.
5. **Treatment Outcome Analysis:** Comparing treatment outcomes across diseases helped identify how recorded outcomes were distributed among different diagnosis categories.
6. **BMI-Based Analysis:** The Normal BMI category had the highest number of recorded Type 2 Diabetes cases. Disease counts varied across BMI categories, but this does not establish a causal relationship.
7. **Clinical Indicator Analysis:** Type 2 Diabetes had the highest average blood glucose level (184.12 mg/dL) and average HbA1c level (8.15%) among the analysed disease groups.
8. **Correlation Analysis:** Blood glucose and HbA1c showed a positive correlation of approximately 0.434, indicating a moderate positive relationship between these two clinical indicators in the dataset.
9. **Monthly Trend Analysis:** July 2024 had the highest overall monthly record count, with 34 records. Type 2 Diabetes reached its highest monthly count in May 2025, with 13 records.

8. Conclusion & Future Scope

The project successfully analysed Indian healthcare patient records using Python, reducing the dataset from 600 to 590 records after duplicate removal. The analysis identified Type 2 Diabetes, Hypertension, and Cardiovascular Disease as the most frequently recorded diseases. Disease patterns were examined across gender, age groups, regions, BMI categories, and treatment outcomes. Type 2 Diabetes showed the highest average blood glucose and HbA1c levels among the analysed disease groups, while correlation analysis revealed a positive relationship between these two indicators. The project demonstrates how data analytics can transform raw healthcare records into meaningful insights.

In the future, the study can be extended by including larger datasets from multiple hospitals and regions to improve the diversity of the analysis. Additional patient attributes, longer observation periods, and regularly updated records could help identify more detailed patterns. Advanced statistical techniques and machine learning models could also be explored to investigate disease-related factors and support early risk assessment, subject to sufficient data and proper validation. Further improvements in data accuracy, missing-value handling, and data privacy practices would help make future healthcare analysis more reliable and useful for research and decision-making.

9. References

1.Kaggle. Healthcare Dataset. Dataset used for disease frequency analysis and healthcare data visualization.

2.Python various libraries were used Pandas ,Numpy ,Matplotlib ,Seaborn ,Folium ,Streamlit, etc. And other tools like Power BI was used for creating Dashboards.

3. ChatGPT. Used for guidance on Folium choropleth maps, Streamlit dashboard development, and improving the grammar and clarity of project documentation

4. Geographic Boundary Data: GitHub repository containing the admin2.shp shapefile and its associated files (.shx, .dbf, and .prj), used for geographic mapping and choropleth map implementation. Add the exact GitHub repository URL from which the files were downloaded.

5.Google Colab and Visual Studio Code: Google Colab was used for data cleaning, analysis, and visualization using Python, while Visual Studio Code was used to develop and run the Streamlit dashboard

6.Dr Sanjay Jain s' Class Notes , Virtual Labs and Learning Materials for Data Analytics. Provided during the course and used as a reference for project concepts and methodology.

10. AI Tool Usage Declaration

☐ No AI tools were used in this project.

☐ AI tools were used, as detailed below:

AI Tool Used	Purpose (e.g., debugging, documentation)	Extent of Use / Section Applied
ChatGPT	To get help with GeoJSON data and Folium choropleth map implementation.	To resolve GeoJSON data issues and understand map implementation.
ChatGPT	To get guidance on developing a Streamlit dashboard as a beginner because it was our first time using Steamlit.	Assistance with dashboard creation, coding, and troubleshooting.
ChatGPT	To improve the clarity and presentation of project documentation.	Assistance with English phrasing, grammar, and sentence structure based on my own notes and explanations.

11. Project Links & Deliverables

GitHub Repository Link:https://github.com/rishugupta23/Disease-Frequency-Analytics

Python Notebook (.ipynb) File Name: Data_Analytics_Project (4).ipynb

PBL Report File Name:DA PBL REPORT

☐ GitHub repository includes code, notebook, *and a PBL report* and README

12. Report Structure Checklist (Section 8 Compliance)

S. No.	Report Section (as per Section 8 format)	Included (✓)
1	Title Page	✓
2	Introduction & Problem Statement	✓
3	Dataset Description & Source	✓
4	Data Cleaning & Preprocessing	✓
5	Exploratory Data Analysis (EDA)	✓
6	Data Visualization & Observations	✓
7	Statistical Analysis (Basic)	✓
8	Key Insights & Interpretation	✓
9	Conclusion & Future Scope	✓
10	References	✓

13. Declaration by Student(s)

I/We hereby declare that this Mini Project Report is our own original work, prepared using publicly available/open datasets. All sources of data, references, and AI tool assistance have been duly acknowledged. I/We understand that any instance of plagiarism in code or report will attract strict academic action, as per Section 15 of the PBL guidelines.

Team Leader Signature: Rishu Gupta

Team Member Signature(s): Amit. Sidhanshu Kumar

Date: 21/10/26

14. For Faculty Use Only — PBL Final Report Evaluation

Evaluation Component B — PBL Final Report Submission / Mid-Term (PBL Based): 10 Marks (Section 10 of PBL Guidelines)

Evaluation Criteria	Max Marks	Marks Awarded
Data Cleaning & Preprocessing	2	1.5
Exploratory Data Analysis (EDA)	2	01
Visualization & Observations	2	02
Statistical Analysis	2	01
Insights, Interpretation & Conclusion	2	1.5
Total (out of 10)	**10**	07

Faculty Remarks: late submission

Faculty Mentor Signature: _____ 22/10/26

Date: _____